EEE1001
ELECTRIC CIRCUITS AND SYSTEMS

Module 4 – Lecture 1

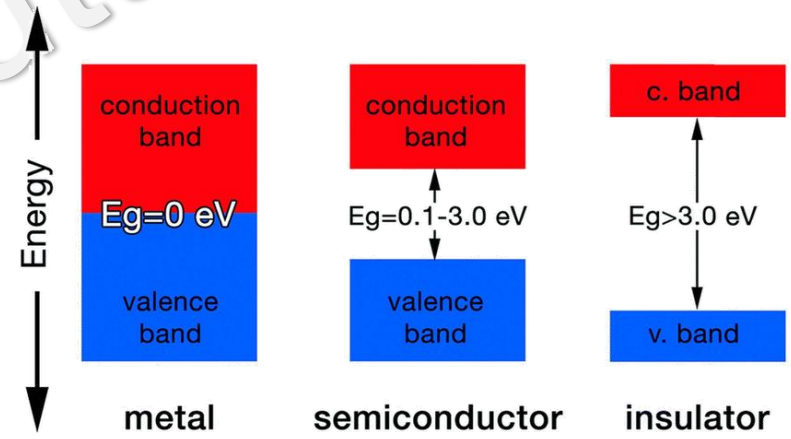
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Semiconductors

- As per atomic theory, matter is composed of exceedingly small particles called atoms. Each atom has a nucleus (containing positively-charged protons and neutral neutrons) and negatively-charged electrons.
- All materials are classified into three categories, namely conductors, semiconductors, and insulators.
- This classification is done based on the band gap (between the valence and conduction bands) of the materials: metals (conductors) have a band gap value = 0 eV; insulators have a band gap value >3.0 eV; and semiconductors have a band gap in the range 0.1-3.0 eV.



- Gold, silver, copper, aluminium, etc., are conducting materials. Conducting materials have a large number of free electrons in their atomic structure which allow flow of current.
- Rubber, ceramic, glass, wood, paper, bakelite, mica, etc. are insulating materials. In these materials no free electrons are available and as such no current should flow through them.
- Substances like germanium, silicon, carbon, etc. are called semiconducting materials. Atoms of these materials binds themselves through sharing of electrons in their outermost shell or orbit. Such bonds are called covalent bonds.
- At absolute zero degree temperature, semiconducting materials behave like insulators as no free electrons are available for conduction. However, with increase of temperature or on application of voltage, some electrons become free electrons by breaking away from their covalent bonds and create a current flow.

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- A semiconductor material, silicon or germanium where the electrons are bound to their respective atoms and are not free to conduct electric current, are called **intrinsic, or pure semiconductors**. When the temperature of the crystal is raised, external energy in the form of heat gets applied to the semiconductor material. This heat energy enables the valence electrons to acquire sufficient energy to break away from the atoms and become free electrons. When an electron leaves to become free, it leaves a vacant space called a hole. For every free electron there will be a corresponding hole produced, which is called an electron–hole pair. When electrons become free, they get attracted and fall into a hole created by another electron. This merging of free electrons and holes is called recombination. The time of creation of a free electron and its falling into a hole is very quick and the time taken is of the order of nanoseconds. Thus, pure silicon or germanium is not of much use in electronics.

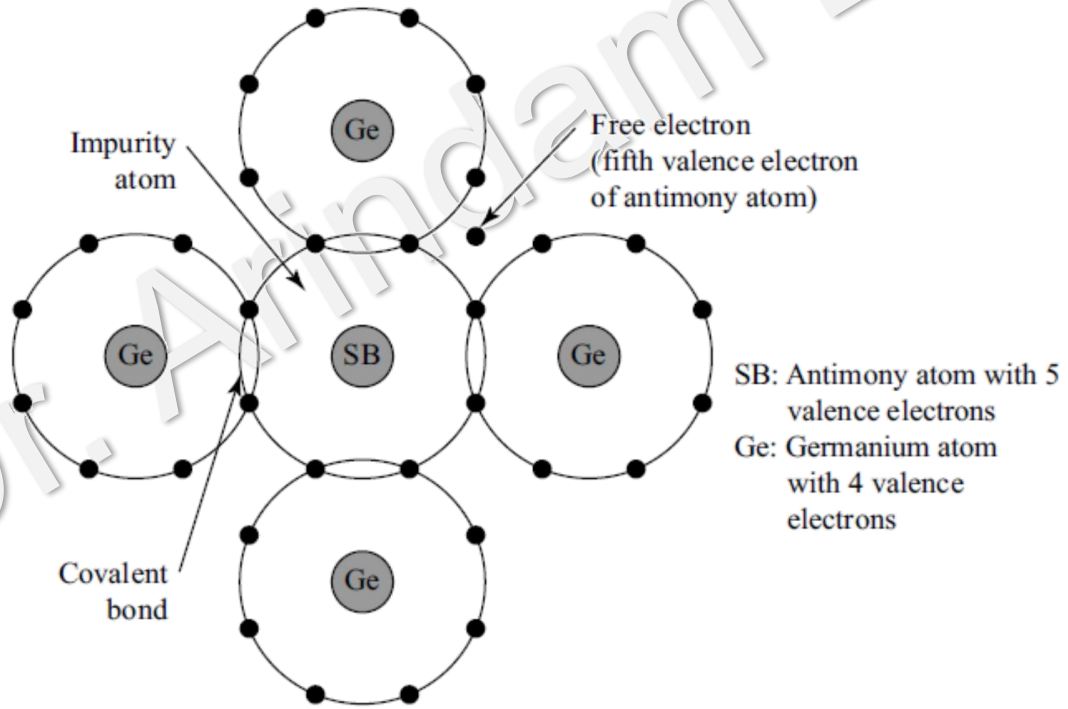
- The conductivity of semiconductor materials can be increased by adding some amount of another material having either three or five valence electrons. Adding such materials with the pure semiconductor material is called **doping**. The material formed by doping is called **extrinsic semiconductors**.

- Extrinsic semiconductors are typically of two types:
    - N-type
    - P-type

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Semiconductors

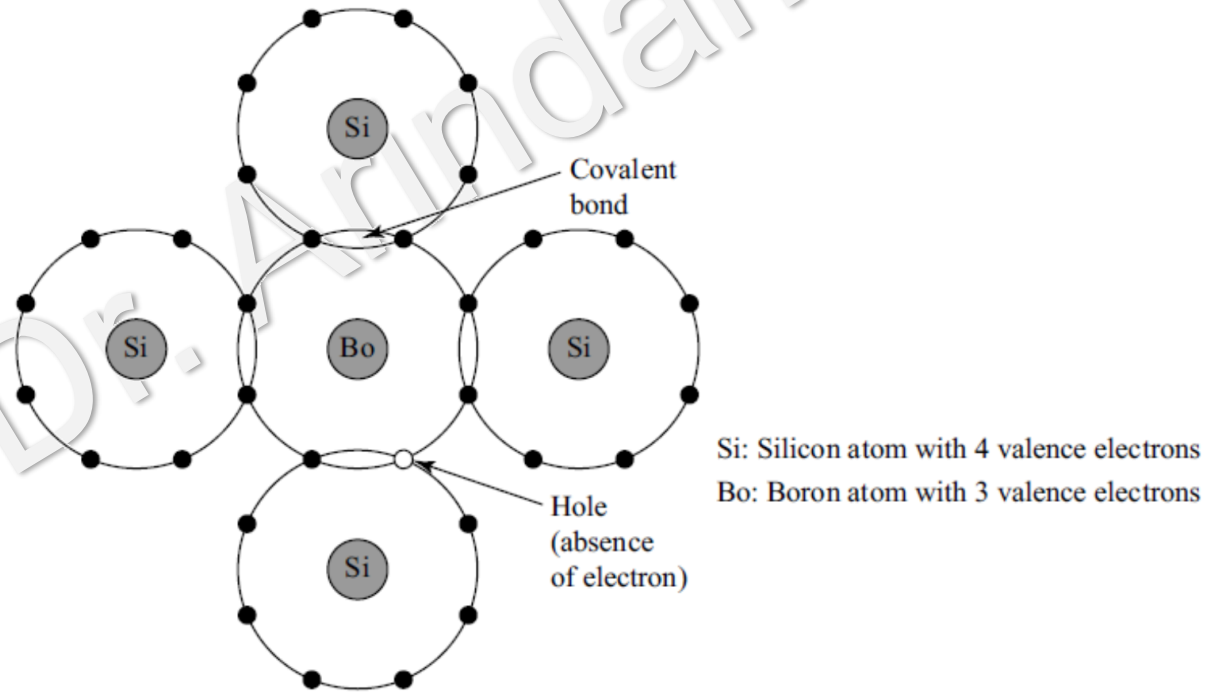
N-Type semiconductor is formed by doping a pure silicon or germanium crystal with a material having five valence electrons. Antimony, arsenic, and phosphorous are pentavalent materials. If arsenic in very small quantity is added to a silicon crystal, four out of five valence electrons will form covalent bonds with silicon atoms with one electron left free. Thus, for each arsenic atom there will be one free electron. Although the percentage of arsenic added is very small, the number of atoms being very large, a huge amount of free electrons will be available in the n-type semiconductor. These electrons being free (not taking part in any covalent bonding) are loosely bound to their parent atom and are free to conduct electricity.



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P-Type material is formed when silicon or germanium crystal is doped with (added with) a small percentage of trivalent impurity material like boron, gallium or indium. When covalent bonds are formed between boron having three valence electrons with silicon having four valence electrons, there will be shortage of one electron in the covalent bonds. This is represented by an empty space in the covalent bonds and is called a hole. There will be one hole corresponding to each of the impurity atoms taking part in forming covalent bonds. This makes seven out of eight positions filled. One position is left vacant which we call a hole. Although the amount of impurity material added is small, the total number of atoms being large produce a large number of holes in the crystal. Here, holes are the majority charge carriers. When temperature is raised, there will be creation of free electrons and holes. These thermally generated electrons will be the minority charge carriers because of their being small in numbers.

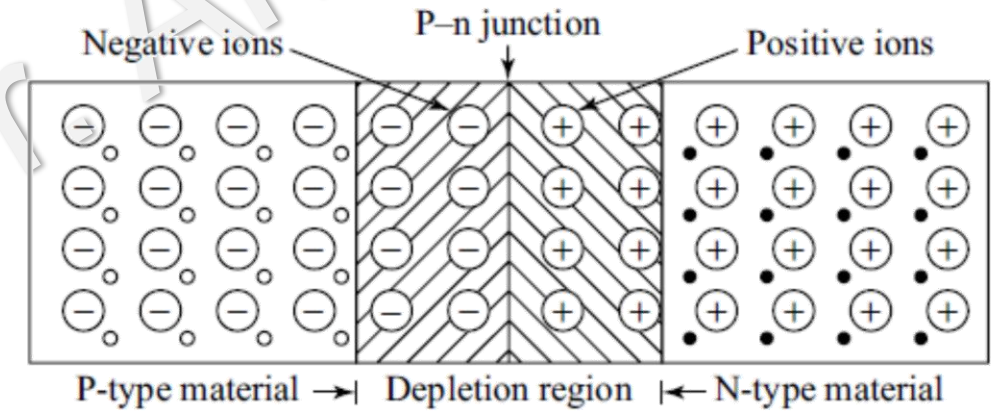


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P-N Junction: When a p-type semiconductor is joined with an n-type semiconductor, a junction called p–n junction is formed. At the p–n junction there will be a tendency of the free electrons from the n-type material to diffuse (move from a high-concentration area to a low-concentration area) into the p-side and combine with a hole nearest to the junction. The free electron crossing over from the n-side to the p-side will leave behind positive immobile ions on the n-side of the junction. The electrons crossing over the junction will occupy the holes in the p-type material making the atoms negatively charged immobile ions. The atoms accepting the negative charge become negatively charged ions which were earlier neutral atoms. Thus, on one side of the junction there is an accumulation of negative ions and on the other side there is accumulation of positive ions. Negative ions created on the p-side close to the junction will acquire a negative voltage and the positive ions created on the n-side close to the junction will acquire a positive voltage. The negative voltage on the p-side will repel further diffusion of electrons from the n-side. The positive voltage on the n-side will repel diffusion of holes from the p-side. When a p–n junction is made there is an initial diffusion of electrons and holes which creates a barrier voltage at the junction, which stops any further diffusion of charge carriers.

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Semiconductors

- This initial diffusion of charge carriers at the junction, and the development resultant barrier voltage take place when a p–n junction is formed during the manufacturing process.

- The barrier voltage depends upon the amount of doping, charge carriers, and the junction temperature. For germanium the barrier voltage is 0.3 V and for silicon the barrier voltage is 0.7 V at room temperature (25 °C).

- The shaded portion on both sides of the p–n junction is having only immobile ions of opposite polarities which creates a potential difference, i.e., barrier voltage. This portion is devoid of any charge carriers. In other words, this region is depleted of any charge carrier, and hence is called the **depletion region**. This layer is very thin, of the order of micrometer.

- Application of some voltage across the p–n junction is called **biasing**.

- Depending on the polarity of biasing the width of the depletion layer will change.

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Forward and Reverse Biasing

(i) A p–n junction is **forward biased** if its p-side is connected to the positive terminal of the supply and n-side is connected to the negative terminal of the supply.

(ii) The depletion layer width gets narrowed down on application of forward voltage.

(iii) The majority charge carriers current is established in a forward-biased p–n junction.

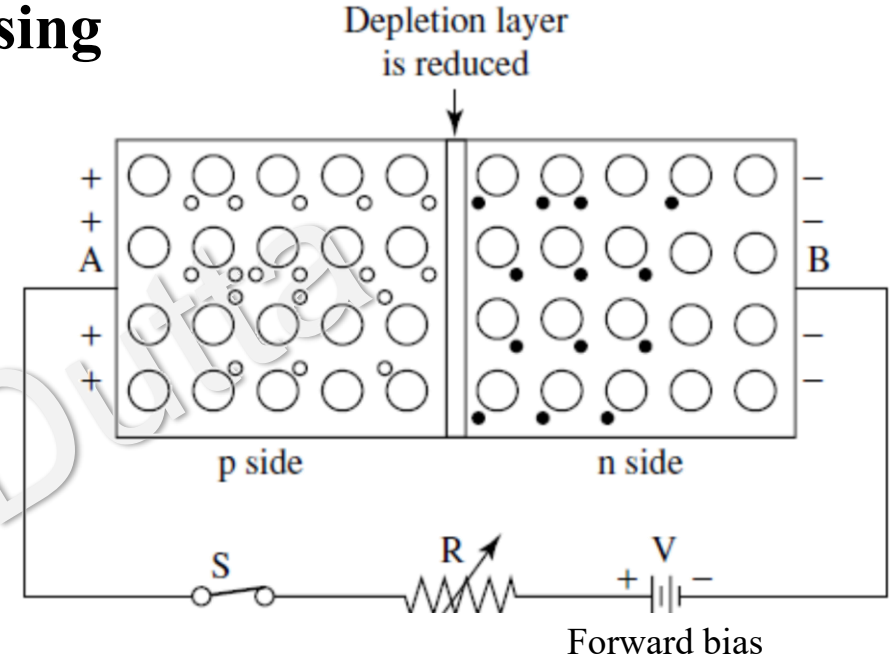
(iv) Barrier potential for germanium is 0.3 V and for silicon is 0.7 V at room temperature. These are the voltage drops across the p–n junction when current flows.

(v) When p-side is connected to the negative terminal and n-side is connected to the positive terminal of the supply, the p–n junction is said to be **reverse biased**.

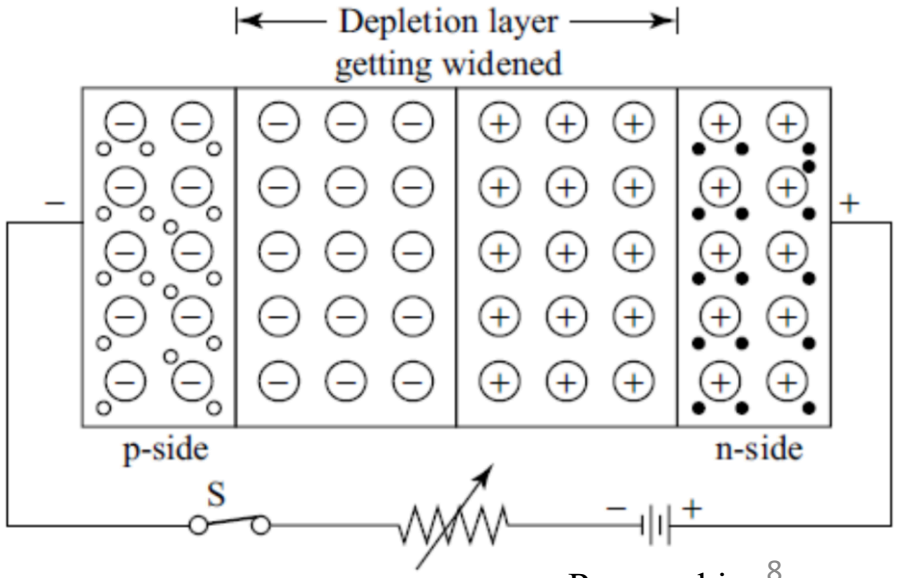
(vi) The width of the depletion layer, and hence the barrier potential increases when the junction is reverse biased.

(vii) A minutely small current flows through a reverse-biased p–n junction due to the minority charge carriers.

(viii) A forward-biased p–n junction offers very small resistance



Forward bias

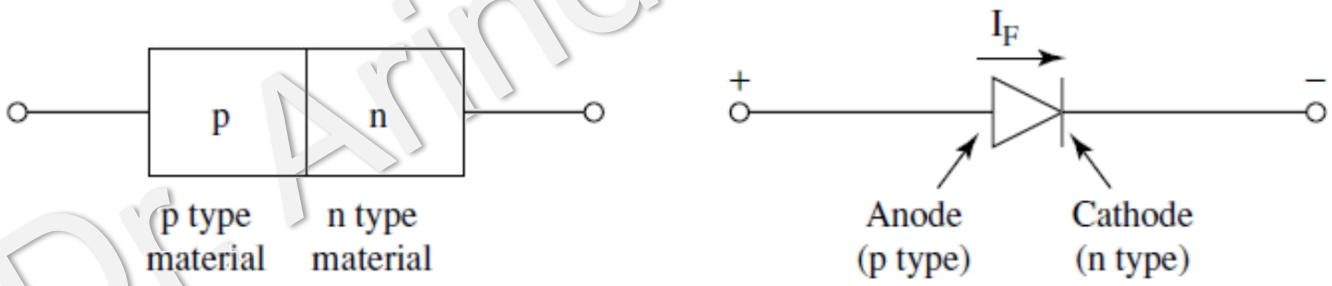


Reverse bias

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Semiconductor Diodes

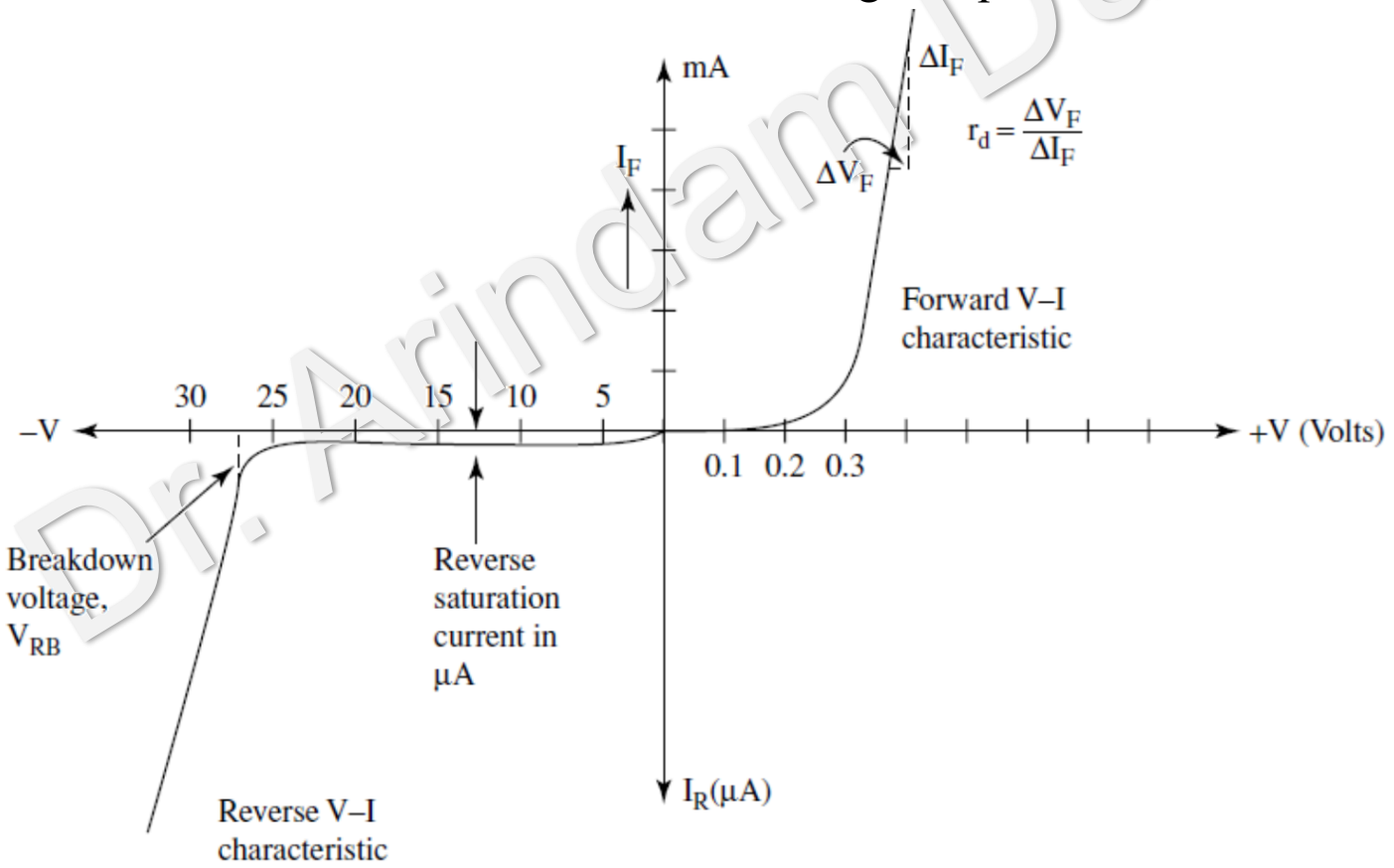
A semiconductor diode is simply a p–n junction which offers very low resistance when forward biased and very high resistance when reverse biased. Diodes are available in different current ratings. Low-current-rated diodes are used in switching circuits as the diode works like a switch allowing current to flow in one direction. A p–n junction with connecting leads on both sides form a p–n junction. The p-side is connected to the positive terminal for forward bias and is called anode. The n-side is connected to the negative terminal for forward bias and is called cathode. A very high forward current or a very high reverse voltage can destroy a diode. That is why the manufacturer, data sheet is to be consulted to note the maximum permissible forward current and maximum permissible reverse voltage. High-current power diodes are available these days which allow large forward current and considerable amount of reverse voltage.



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Semiconductor Diodes

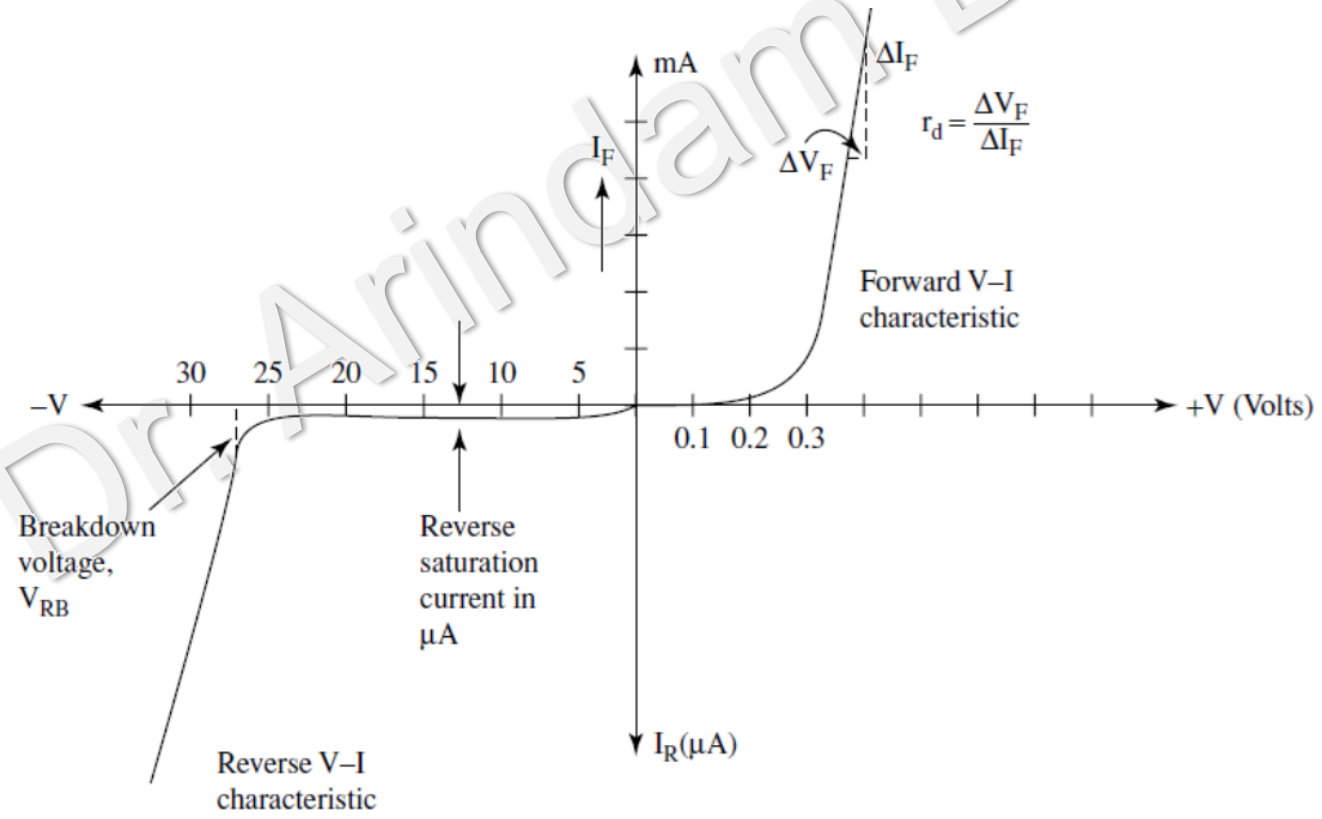
When a p–n junction diode is connected to a source of supply in such a way that it is forward biased, the relationship between the voltage applied and current flowing will give us a forward V–I characteristic. When the applied voltage is gradually increased, at a small value of forward voltage the forward current is negligibly small. At a voltage near 0.3 V, the current suddenly increases. This voltage at which the forward current starts increasing is called the **cut-in voltage** of the diode. The voltage drop across the diode while it is conducting remains almost constant. For the germanium semiconductor diode, the forward voltage drop is 0.3 V and for the silicon diode, the forward voltage drop is 0.7 V.



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Semiconductor Diodes

For determining the reverse characteristic, the supply connection has to be reversed. Under the reverse-biased condition, the junction resistance is very high and ideally no current should flow. But due to minority charge carriers, a negligibly small current of the order of μA will flow. This current is also called leakage current of the diode. It gets saturated to its initial value of a few μA or even less than that. Increase of negative biasing, i.e., increase of negative voltage across the diode does not increase this reverse current. However, if the reverse voltage is increased to a large value, at one stage, the p–n junction will break down with a sudden rise in reverse current. The reverse voltage at which the diode breaks down and a large reverse current starts flowing is called the **breakdown voltage**. At this reverse breakdown voltage, current continues to increase.



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Semiconductor Diodes

A diode is specified in terms of certain parameters. These are as follows:

(i) Forward Voltage drop, V_F
(ii) Reverse Breakdown Voltage, V_{RB}
(iii) Reverse saturation current, I_R
(iv) Dynamic resistance, r_d
(v) Maximum forward current, I_{FM}

The dynamic resistance, r_d is calculated from the slope of the forward V–I characteristic

$$r_d = \frac{\Delta V_F}{\Delta I_F} \Omega$$

The values of these parameters are normally provided by the manufacturers in their specification sheet. Diodes are available in low-, medium-, and high-current ratings. Diodes of low-current ratings are used in electronic switching circuits, i.e., they work as switches. Their forward current ranges from a few mA to a maximum of 100 mA. The safe reverse bias that can be applied is around 75 V. The reverse saturation current is very small, usually less than a micro-ampere. Medium current diodes have a maximum current rating of 400 mA and reverse voltage of about 200 V. High-current diodes are also called power diodes. They are rated for high current and high reverse voltage ratings. Metal heat sinks are used for dissipation of heat produced in a diode when it is conducting. In addition to their use in switching circuits, diodes are used in rectifier circuits for half-wave and full-wave rectification.

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Zener Diode

When a diode is reverse based, only a minutely small current called saturated current flows (ideally no current should flow). If the reverse voltage is increased continuously, the junction breaks down and suddenly a large reverse current flows. This reverse current is limited by connecting a suitable series resistance so that excessive heat produced due to heavy current flow may not burn the diode. If the reverse breakdown current is limited to the current-carrying capacity of the diode, it can be operated under reverse breakdown condition. The V–I characteristic of the diode under the reverse-biased condition can be made dropping down almost vertically by proper doping of the semiconductor material. A diode with a very sharp breakdown voltage is called a Zener diode. Diodes designed to operate under the reverse breakdown condition, maintain a fairly constant voltage over a wide range of current levels. When the reverse voltage is reduced below the breakdown voltage, the current level returns to the very low saturation current level.

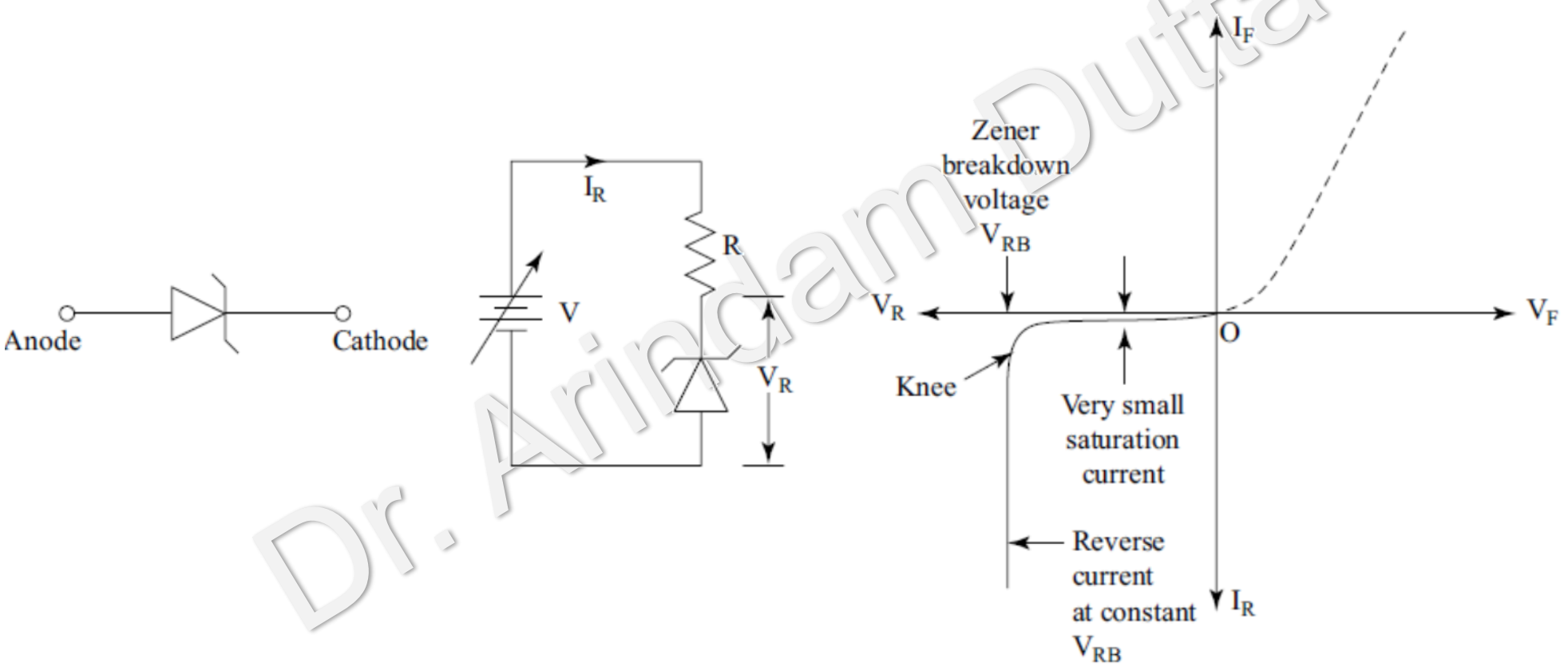
There are two ways that breakdown of a zener diode may occur. One is called **zener breakdown** and the other is called **avalanche breakdown**. If the depletion layer of a diode is narrow and we apply a reverse voltage, the voltage per unit of width of the depletion layer becomes high. This establishes a strong electric field intensity which causes electrons to break away from their parent atoms. Thus, a depletion layer which was insulating in nature, becomes a conducting path. This kind of breakdown due to the creation of a strong electric field intensity, i.e., V/μm is called **zener breakdown**.

If the width of the depletion layer is wide for a zener breakdown, a sufficient reverse voltage may provide the free electrons (minority carriers causing saturation current) to gain sufficient energy to knockout electrons from the atoms of the semiconductor in the depletion region. This is called ionization by collision. The breakdown occurring this way is called **avalanche breakdown**.

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Zener Diode

Zener breakdown occurs at a voltage less than 5 V and avalanche breakdown voltage is higher than 5 V. The forward V–I characteristic of a zener diode is the same as an ordinary diode.



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Zener Diode

The dynamic resistance of the zener diode, also known as the zener resistance, is similar to the dynamic resistance of a forward-biased diode. Zener resistance, R_Z is given by

$$R_z = \frac{\Delta V_z}{\Delta I_z}$$

The value of R_Z is to be small so that it will indicate a steep curve where due to a small change in voltage, ΔV_Z a large change in current, ΔI_Z takes place. The value of zener resistance varies with the current. It decreases with increase in current.

Zener diodes are often used as a **voltage regulator or stabilizer**, and as **reference voltage** in electronic circuits.

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Module 4 – Lecture 2

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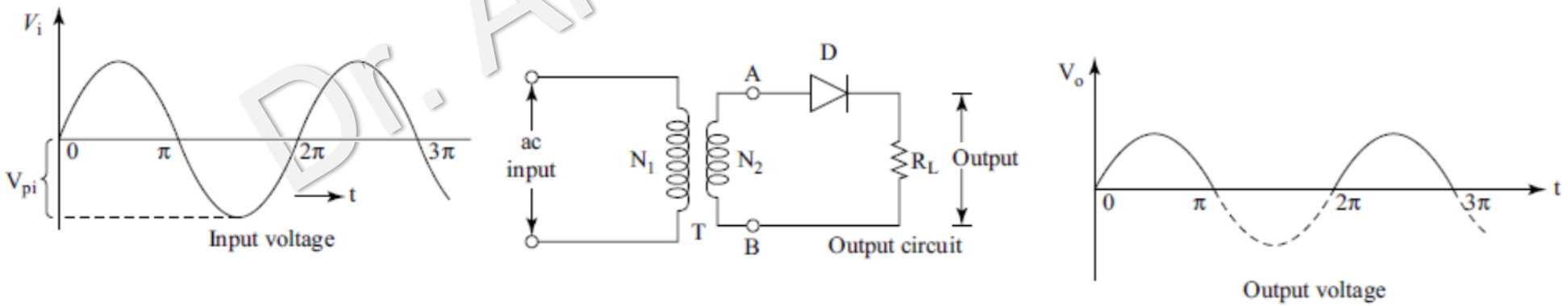
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Rectifiers

A rectifier is a device that converts ac supply into dc using diodes. Rectification can be done by halfwave or full-wave rectification circuits.

A **half-wave rectifier** circuit consisting diode, and a load represented by a resistor has been shown in the figure. The input and output wave shapes have been shown of a transformer. The diode is forward biased during the positive half cycle of the applied voltage and reverse biased during the negative half cycle.

During the positive half cycle, i.e., from time 0 to π, voltage is positive, and hence terminal A of the transformer is positive. Diode is forward biased, and hence current will flow through the diode and through the load resistance, R_L. If a step down transformer is used, the magnitude of output voltage will be reduced. If however, step down of voltage is not required, a transformer having an equal turn ratio, i.e., N_2/N_1 will be used. The function of the transformer will be to electrically isolate the dc output circuit from the input circuit.



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Rectifiers

During the negative half cycle of the input voltage wave, the terminal A of the transformer will be negative and terminal B will be positive. The diode will be reverse biased and virtually no current will flow through the diode and the load. This sequence of allowing the positive half cycle of current through the diode and blocking the negative half cycle will continue for each and every cycle of power supply. As a result, a half-wave rectified current will flow through the output circuit. The voltage drop across the load resistance, R_L will be the output voltage, V_O which will be a half-wave rectified voltage. When the diode is reverse biased, the maximum or peak voltage of the negative half cycle of the input will be appearing across the diode terminals. This is the peak of the reverse voltage or **peak-inverse voltage (PIV)** which is getting applied to the diode in every alternate half cycles.

A diode works as a closed switch during the positive half cycle of the input voltage and works as an open switch during the negative half cycle. The output voltage appears across the load during the positive half cycle only. The load voltage and load current although positive all the time (i.e., unidirectional) are fluctuating dc as its magnitudes are changing. Our aim will be to obtain steady dc at the output. **The average and rms values of this fluctuating dc can be calculated as I_m/π and $I_m/2$, respectively.**

The performance parameters of a half-wave rectifier output are calculated in terms of the output dc current (i.e., the average value of the rectified wave), the rms value of the output current, the output voltage, ripple factor, PIV, etc.

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Rectifiers

Ripple Factor: The output of a half-wave rectifier is a pulsating dc. If we analyse, we will see that it has a steady dc component and an ac component. The ac component is called the ripple. Ripple factor is defined as the ratio of the rms value of ac component to the value of dc component. The ripple factor indicates the level of fluctuation of the output voltage from its steady value. Ripple is an undesired effect and should be minimized. The ripple factor, r for a half-wave rectifier is calculated as

$$\text{Ripple Factor, } r = \frac{\text{RMS value of ac component}}{\text{value of dc component}} = \frac{I_{ac}}{I_{dc}}$$

Also, $I_{rms} = \sqrt{I_{dc}^2 + I_{ac}^2}$ or, $I_{ac} = \sqrt{I_{rms}^2 - I_{dc}^2}$

where I_{ac} is the rms value of the ac component of the output current, I_{dc} is the dc component of the output current, and I_{rms} is the rms value of the output current. Therefore,

$$r = \frac{\sqrt{I_{rms}^2 - I_{dc}^2}}{I_{dc}} = \sqrt{\left(\frac{I_{rms}}{I_{dc}}\right)^2 - 1}$$

Considering $I_{rms} = I_m/2$ and $I_{dc} = I_m/\pi$, $r = \sqrt{\left[\frac{I_m/2}{I_m/\pi}\right]^2 - 1} = 1.21$

Output voltage, V_{dc} across the load is $V_{dc} = I_{dc} R_L = \frac{I_m}{\pi} R_L = \frac{V_m}{\pi} = 0.318\ V_m$

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Rectifiers

Rectifier efficiency: It is calculated as the ratio of output power to input power.

DC output power, $P_{dc} = I_{dc}^2 R_L = \left(\frac{I_m}{\pi}\right)^2 R_L$

AC input power, P_{ac} = Power dissipated in diode junction + Power dissipated in the load $= I_{rms}^2 R_f + I_{rms}^2 R_L$

The forward resistance, R_f of the diode is very small, and hence $I^2_{rms}R_f$ can be neglected in comparison with $I^2_{rms}R_L$.

Therefore, rectifier efficiency $= \frac{P_{dc}}{P_{ac}} = \frac{I_m^2}{\pi^2} RL \div I_{rms}^2 R_L$

$$= \frac{I_m^2 RL}{\pi^2 I_{rms}^2 RL} = \frac{I_m^2}{\pi^2 \cdot \left(\frac{I_m}{2}\right)^2}$$

$$= \frac{4}{\pi^2} = 0.406 = 40.6 \text{ per cent}$$

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This value of efficiency is considered low and ripple factor is considered very high. A filter circuit has to be used to minimize the ripples.

Rectifiers

Peak inverse voltage (PIV): As mentioned earlier, PIV is the maximum value of reverse voltage that appears across the diode when it gets reverse biased. Here

$$\mathbf{PIV = V_m}$$

Voltage Regulation of the rectifier is calculated using the relation,

$$\text{Voltage regulation} = \frac{V_{dc} \text{ at no load} - V_{dc} \text{ on full load}}{V_{dc} \text{ on full load}}$$

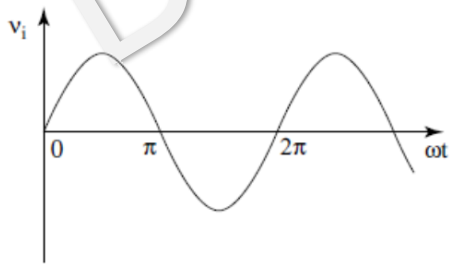
The difference between no-load voltage and full-load voltage is the voltage drop in the transformer winding and across the diode. The value of voltage regulation, which is generally expressed in percentage should be low.

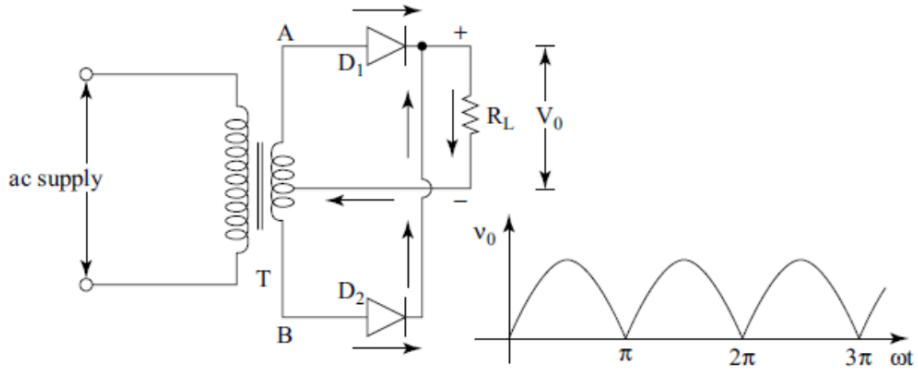
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Rectifiers

Full-wave rectifiers can be made using two diodes and a centre-tapped transformer. Full-wave rectifiers are also made using a two-winding transformer and four diodes. Such rectifiers are called bridge rectifiers.

Two-diode full-wave rectifier: here ac input voltage is supplied from the secondary of a centre-tapped transformer. The circuit consists of the transformer, two diodes and the load resistance. The circuit is essentially the summation of two half-wave rectifiers. For the positive half cycle of the input voltage, diode D1 will conduct. This is because terminal A is positive and the diode D1 is forward biased. As terminal B is negative, diode D2 will not conduct. In the negative half cycle, terminal B is positive and terminal A is negative. Hence, diode D2 will conduct and diode D1 will be reverse biased. This way in each half cycle one of the two diodes will conduct and current will flow through the load resistance, R_L. The output current and the output voltage across the load will be a full-wave rectified current and voltage, respectively. The output wave form is a series of consecutive positive half cycles of sinusoidal wave form. The current through the load resistance is unidirectional but its magnitude is fluctuating. The PIV is the maximum voltage that would appear across a diode when it is reverse biased. Here, when D1 is conducting, D2 is reverse biased and vice versa. When D1 is conducting, the voltage that would appear across diode D2 is the sum of the voltage across the lower half of the transformer secondary winding and the voltage appearing across the load. PIV of the diode is equal to $2V_m$.



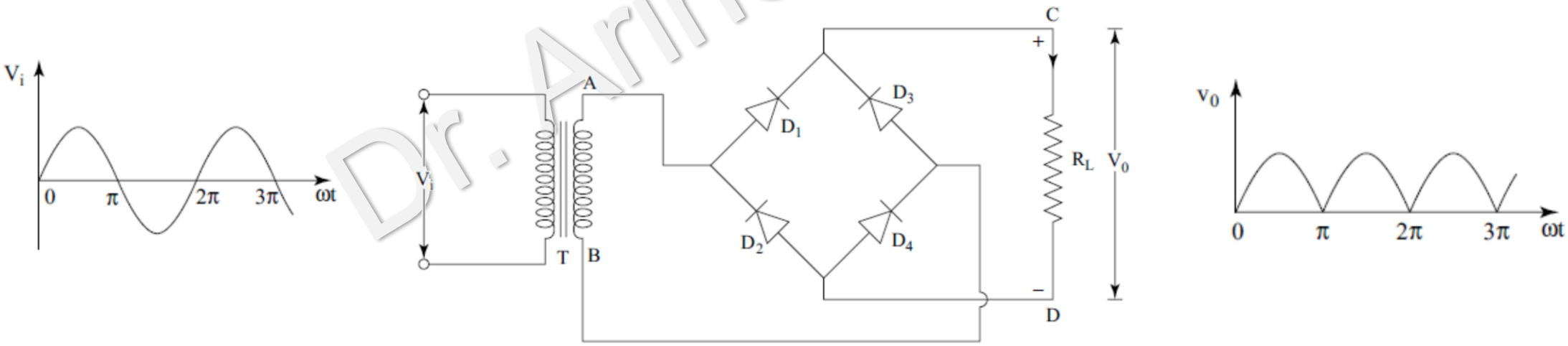


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Rectifiers

A **bridge rectifier** circuit uses four diodes connected in the form of a bridge. During the positive half cycle of the input voltage, terminal A of the transformer is positive. Current will flow from the positive terminal through diode D1, load R_L and diode D4, and back to the negative terminal, B of the transformer. The direction of current through the load has been shown to be from C to D, i.e., from top to bottom. The polarities of the load terminals have been shown. During this period diodes D2 and D3 are reverse biased. During the negative half cycle of the input voltage diodes D3 and D2 are forward biased while diodes D1 and D4 are reverse biased. Current through the load will flow in the same direction, i.e., from terminal C to D. During both positive and negative half cycles of the input voltage, current will pass through the load in the same direction. The output voltage wave shape is a series of positive half cycles of the sinusoidal voltage. This is dc output but having a varying magnitude. The transformer provides isolation of the dc output from the supply ac input.

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Rectifiers

The average value or dc value of load current (I_{dc}) of a full-wave rectifier will be the same if calculated for a period 0 to π or 0 to 2π and turns out to be $2I_m/\pi$. The rms value of the load current (I_{rms}) is $I_m/\sqrt{2}$.

Output voltage, V_{dc}

$$V_{dc} = I_{dc} R_L$$

$$= \frac{2\ I_m}{\pi} R_L$$

$$= \frac{2\ R_L}{\pi} \frac{V_m}{(R_L + 2R_F + R_2)}$$

where R_L is the load resistance, R_F is the forward resistance of the diode, and R_2 is the secondary winding resistance of the transformer.

Rectifier efficiency, η

$$\eta = \frac{\text{dc power output, } P_{dc}}{\text{ac power input, } P_{ac}}$$

$$= \frac{I_{dc}^2 R_L}{I_{rms}^2 (R_L + 2R_F + R_2)}$$

(Two diodes being in series)

$$= \frac{(2I_m)^2 R_L}{\pi^2 I_{rms}^2 (R_L + 2R_F + R_2)} \qquad = \frac{(2\sqrt{2}\ I_{rms})^2 R_L}{\pi^2 I_{rms}^2 (R_L + 2R_F + R_2)} \qquad = \frac{8\ I_{rms}^2}{\pi^2 I_{rms}^2 \left(1 + \frac{2R_F + R_2}{R_L}\right)}$$

$$= 0.812 \text{ since } (2R_F + R_2) << R_L$$

$$= 81.2 \text{ per cent}$$

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Rectifiers

$$\text{Ripple factor, } r = \frac{\text{RMS value of ac component}}{\text{dc component}}$$

$$= \sqrt{\left(\frac{I_{rms}}{I_{dc}}\right)^2 - 1}$$

Substituting, $I_{rms} = \frac{I_m}{\sqrt{2}} =$ and $I_{dc} = \frac{2I_m}{\pi}$

$$r = \sqrt{\left(\frac{I_m \pi}{\sqrt{2}\ 2\ I_m}\right)^2 - 1}$$

$$= \sqrt{\frac{\pi^2}{8} - 1}$$

$$= 0.48$$

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We had earlier calculated the ripple factor for a half-wave rectifier as 1.21. For a full-wave rectifier, the ripple factor is reduced to 0.48. This indicates that the fluctuation of dc output is reduced. **PIV for a bridge rectifier = V_m.**

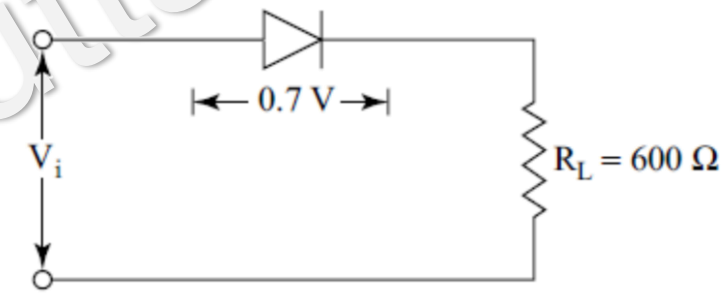
Rectifiers

Parameters	Half-wave rectifier	Full-wave rectifier with centre-tapped transformer	Bridge rectifier
Number of diodes regd.	1	2	4
DC load current, I_{dc}	$\frac{I_m}{\pi}$	$\frac{2I_m}{\pi}$	$\frac{2I_m}{\pi}$
RMS value or load current I_{rms}	$\frac{I_m}{2}$	$\frac{I_m}{\sqrt{2}}$	$\frac{I_m}{\sqrt{2}}$
DC output, P_{dc}	$I_{dc}^2 R_L = \frac{I_m^2}{\pi^2} R_L$	$\frac{4 I_m^2}{\pi^2} R_L$	$\frac{4 I_m^2}{\pi^2} R_L$
AC input, P_{ac}	$I_{rms}^2 (R_L + R_F + R_2) = \frac{I_m^2}{4}(R_L + R_F + R_2)$	$I_{rms}^2 (R_L + R_F + R_2) = \frac{I_m^2}{2}(R_L + R_F + R_2)$	$I_{rms}^2 (R_L + 2R_F + R_2) = \frac{I_m^2}{2}(R_L + 2R_F + R_2)$
Maximum rectification efficiency	40%	81.2%	81.2%
Ripple factor	1.21	0.48	0.48
PIV	V_m	$2V_m$	V_m
Ripple frequency	Fr = f	Fr = 2f	Fr = 2f
Centre tap transformer	Not required	Required	Not required
Transformer utilization factor	28.7%	69.2%	81.2%

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Problem

Q: A half-wave diode rectifier has a forward voltage drop, i.e., voltage drop across the diode when conducting is 0.7 V. The load resistance is 600 Ω. The rms value of the ac input is 28.87 V. Calculate I_{dc}, I_{rms}, PIV, and form factor.



$$V_i(\text{RMS}) = 28.27\ \text{V}$$

$$V_i(\text{max}) = \sqrt{2}\, V_i(\text{RMS})$$

$$= 1.414 \times 28.27$$

i.e., $V_m = 40\ \text{V}$

$$\text{PIV} = V_m = 40\ \text{V}$$

$$I_{dc} = \frac{I_m}{\pi}; \qquad I_m = \frac{V_m - 0.7}{R_L} = \frac{40 - 0.7}{600} = \frac{39.3}{600}\ \text{A}$$

$$I_{dc} = \frac{39.3}{600 \times \pi} = 0.0208\ \text{A} = 20.8\ \text{mA}$$

$$I_{rms} = \frac{I_m}{2}\ \frac{39.3}{600 \times 2} = 0.0327\ \text{A} = 32.7\ \text{mA}$$

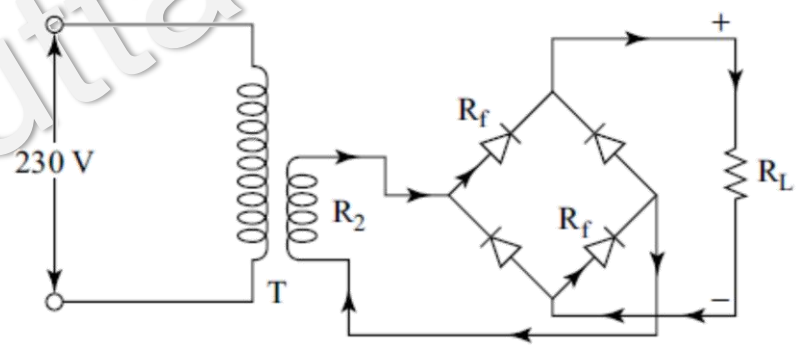
$$\text{Form factor} = \frac{\text{RMS value}}{\text{Average value}} = \frac{I_{rms}}{I_{dc}} = \frac{32.7}{20.8}$$

$$= 1.57$$

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Problem

Q: The input to a bridge rectifier is through a step-down transformer of turn ratio 10:1. The supply voltage is 230 V at 50 Hz. The load resistance is 1.2 kΩ, secondary winding resistance of the transformer is 4 Ω, and diode forward resistance is 2 Ω. Calculate the efficiency of the bridge rectifier.



Given V_i (RMS) = 230 V

$R_F = 2\ \Omega,\ R_2 = 4\ \Omega,\ R_L = 1200\ \Omega$

The RMS value of the emf in transformer secondary

$$V_s(\text{RMS}) = 230\left(\frac{N_2}{N_1}\right)$$

$$= 230\left(\frac{1}{10}\right)$$

$$= 23\ \text{V}$$

Peak secondary voltage,

$$V_m = \sqrt{2}\, V_s(\text{RMS})$$

$$= \sqrt{2} \times 23$$

$$= 32.5\ \text{V}$$

Current through the load will flow from the transformer secondary via two diodes. Therefore, following the current path during the positive half cycle:

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$$I_m = \frac{V_m}{R_L + 2\,R_F + R_2} = \frac{32.5}{1200 + 4 + 4} = 26.8 \text{ mA}$$

For a bridge rectifier,

$$I_{dc} = \frac{2I_m}{\pi} = \frac{2 \times 26.8}{3.14} = 17 \text{ mA}$$

DC power output, $P_{dc} = I_{dc}^2 R_L = (17 \times 10^{-3})^2 \times 1200$

$$= 346.8 \text{ mW}$$

AC power input,

$$P_{ac} = (I_{rms})^2 \, (R_L + 2\,R_F + R_2)$$

$$= \left(\frac{I_m}{\sqrt{2}}\right)^2 (R_L + 2\,R_F + R_2)$$

$$= \left(\frac{26.8 \times 10^{-3}}{2}\right)^2 \times (1200 + 2 \times 2 + 4)$$

$$= 432 \text{ mW}$$

$$\text{Rectifier efficiency, } \eta = \frac{P_{dc}}{P_{ac}} \times 100 = \frac{346.8 \times 100}{432} = 80 \text{ per cent.}$$

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EEE1001
ELECTRIC CIRCUITS AND SYSTEMS

Module 4 – Lecture 3

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Bipolar Junction Transistors

- Transistors are used to amplify electrical signals. The word transistor is the short form of the word “transfer resistor”. The signal amplification is achieved by transferring the signal from a region of low resistance to a region of high resistance.

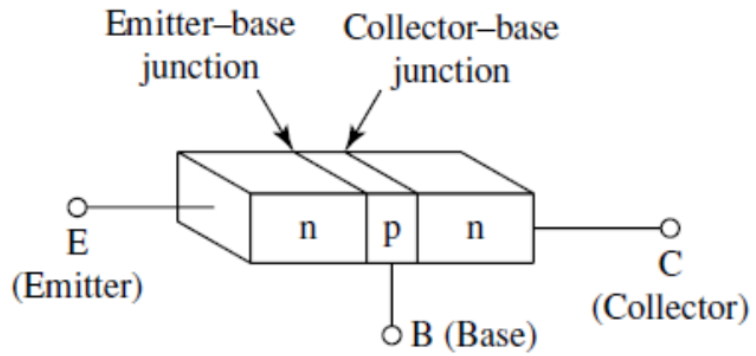
- A bipolar junction transistor (BJT) has three layers of semiconductor material. These layers are arranged either in an n–p–n sequence or in a p–n–p sequence. In an n–p–n transistor, a p-type semiconductor material is sandwiched between two n-type materials. In a p–n–p transistor, an n-type semiconductor material is sandwiched between two p-type materials. A transistor, in general, has two p–n junctions connected back to back. The central layer is called the base, one of the outer layers is called the emitter, and the other is called the collector.

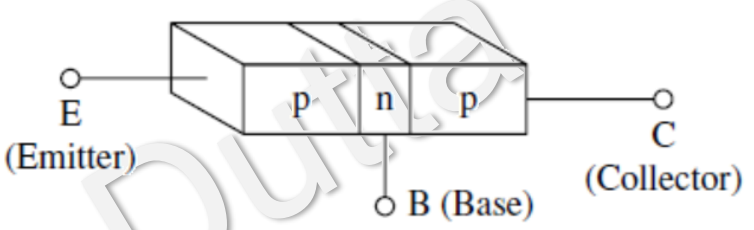
- The basic principle of transistor operation is that a small current in the base region can control a much larger current flow through the transistor, i.e., from the emitter to the collector. A transistor can be used as current amplification or a voltage amplification device. Since a transistor combines two junction diodes, it works on the basis of p–n junction theory.

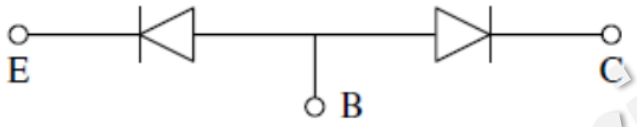
- The symbol for an n–p–n or a p–n–p transistor is the same except for the direction of the arrow head. The arrow head has to be shown from p terminal to n terminal between the emitter and the base. The emitter of a transistor is heavily doped. The base is lightly doped while the collector is less heavily doped than the emitter.

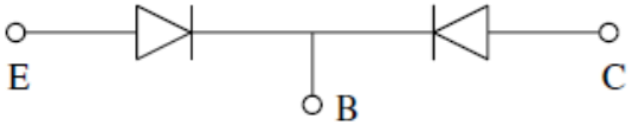
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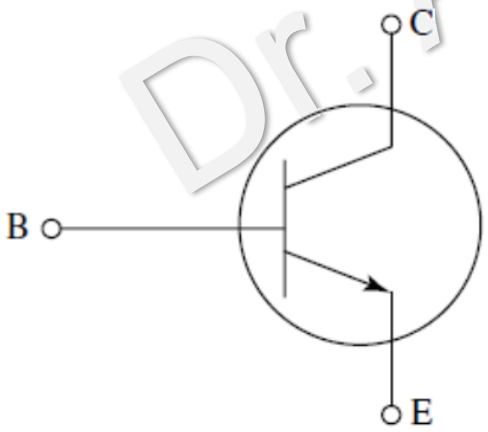
Bipolar Junction Transistors

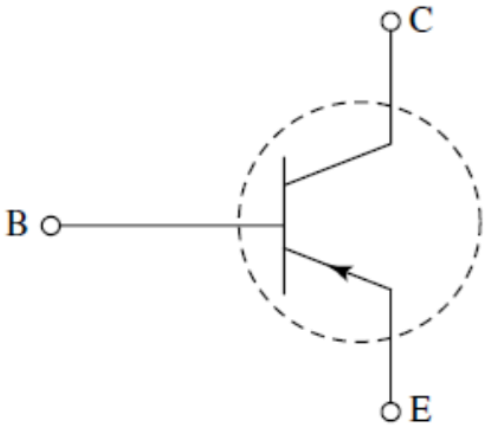












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Bipolar Junction Transistors

- BJTs can operate in three regions namely cut-off, active, and saturation regions.
- To operate the transistor in the desired region we have to apply the external DC voltages of correct polarity and magnitude to the two junctions of the transistor.

Region of operation	Base-Emitter Junction	Collector-Base Junction	Application
Cut-off	Reverse biased	Reverse biased	As an open switch (OFF)
Active	Forward biased	Reverse biased	As an amplifier
Saturation	Forward biased	Forward biased	As a closed switch (ON)

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Bipolar Junction Transistors

Working of n-p-n transistor: For a transistor to work, it has to be biased by applying external voltage supply with proper polarity. For operation in the **active region**, a transistor's emitter–base junction must be forward biased while the collector base region reverse biased. The forward bias of the EB junction reduces the width of the depletion region and the barrier voltage gets reduced. The reverse bias of the CB junction increases the width of the depletion region and the barrier voltage gets increased.

Since the EB junction is forward biased, electrons form the majority charge carriers and would flow from the emitter to the base region. Since the base is lightly doped, there will be a smaller number of holes present there. About 2% of the electrons from the emitter recombine with some holes that are present in base region, while a smaller number of holes are injected from base to emitter. This requires a continuous supply of holes from the external base terminal, thus generating a small base current.

The reverse bias of the CB junction causes expansion of the depletion layer. The width of the base region is thinner than the collector region. Therefore, the depletion region will penetrate deep into the base region. The electrons from the emitter region will arrive near the CB depletion region. Due to large CB bias voltage, electrons will be pulled across the CB junction by the positive terminal of the collector. The collector thus collects the 98% of the electrons emitted by the emitter. As the reverse bias voltage is increased, the base gets thinner, reducing the likelihood of charge recombination, and further increasing the collector current. This phenomenon is known as the **early effect**.

The quantity of charge carriers crossing the emitter to the base is controlled by the base–emitter bias voltage. Thus, it can be said that emitter and collector current levels can be controlled by the base–emitter bias voltage. For a silicon transistor, substantial current will start flowing only when the bias voltage V_{BE} is 0.7 V and for the germanium transistor V_{BE} is 0.3 V.

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Bipolar Junction Transistors

To sum up an **n-p-n transistor**:

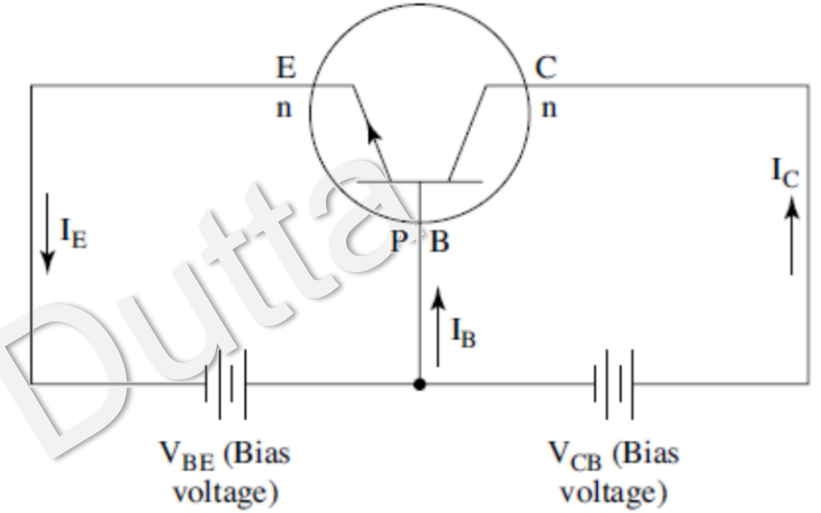
(i) The outer layers of n–p–n transistors are called emitter and collector, respectively, the central layer is called the base.

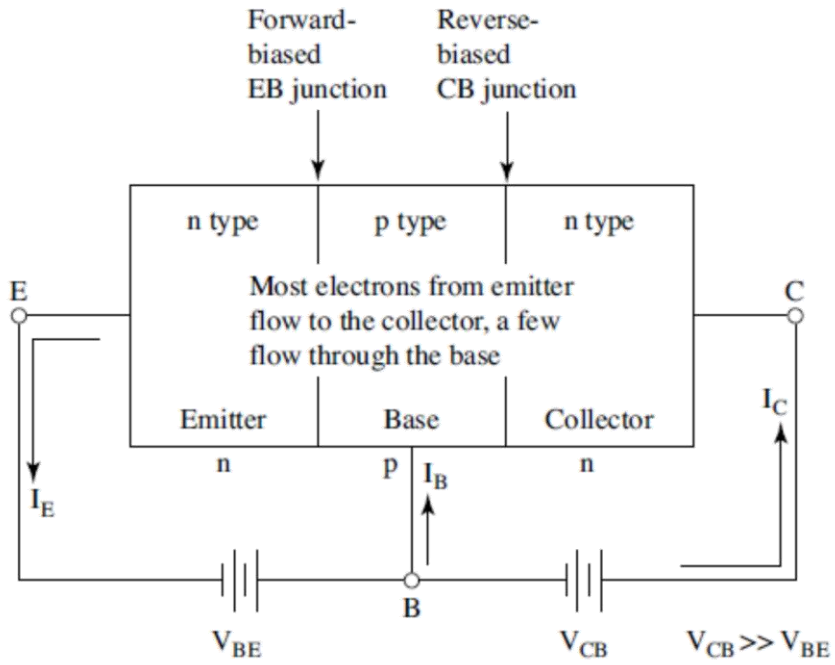
(ii) The base–emitter junction, EB is forward biased and the collector–base junction is reverse biased, and V_{CB} is higher than V_{BE}.

(iii) The base section is made very thin and is lightly doped so that majority of the charge carriers (electrons in n–p–n transistor) can move from the emitter to the collector.

(iv) Most charge carriers flow from emitter to collector and only a small percentage flow through the base material.

(v) Variation of base–emitter voltage changes base, emitter, and collector currents.





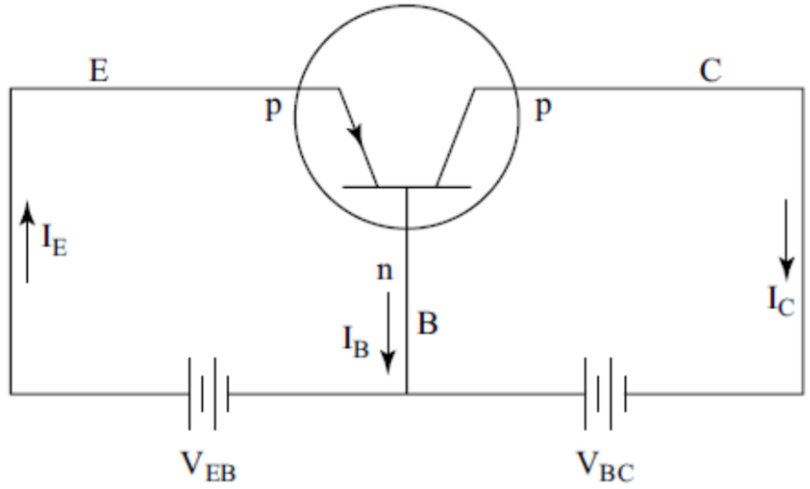
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Bipolar Junction Transistors

Working of p-n-p transistor: They work the same way as n–p–n transistors except that in p–n–p transistors the majority charge carriers are holes. The biasing is same as that of a n–p–n transistor. The emitter-base junction is forward biased and the collector–base junction is reverse biased. The majority charge carriers from the emitter are the holes.

As the base is thin and lightly doped only a small percentage of holes emanating from the emitter recombine with electrons in the base region. The rest of the holes, which are nearly 98%, cross the base–collector barrier potential, because they are attracted by the negative terminal of the base–collector bias voltage. Thus, holes are emitted from the p-type emitter across the forward-biased emitter–base junction and only a few of them find electrons in the base region to recombine with. Most of the holes get attracted to the collector side by the reverse-biased collector–base junction.

By varying the forward bias voltage at the base–emitter junction, we can control the large emitter and collector current through small variations of base current.



Both n–p–n and p–n–p type of transistors are called bipolar junction transistors, or simply BJTs because the charge carriers for both types of transistors are electrons, and holes, although for n–p–n transistors the majority charge carriers are electrons and for p–n–p transistors the majority charge carriers are the holes.

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Bipolar Junction Transistors

The ratio of the collector current to the emitter current of a transistor is called emitter to collector current gain (α_{dc}) which is typically between 0.95 and 0.99.

Here, $I_E = I_C + I_B$

Also, $I_C = \alpha_{dc} I_E$

So, $I_B = (1 - \alpha_{dc}) I_E$

As the collector–base junction is reverse biased, a small reverse saturation current flows across the junction due to minority charge carriers which are very small in number, and can be neglected.

Thus, $I_C = \alpha_{dc} I_E = \alpha_{dc} (I_C + I_B)$

$$I_C (1 - \alpha_{dc}) = \alpha_{dc} I_B$$

$$I_C = \frac{\alpha_{dc}}{1 - \alpha_{dc}} I_B$$

$$I_c = \beta_{dc} I_B$$

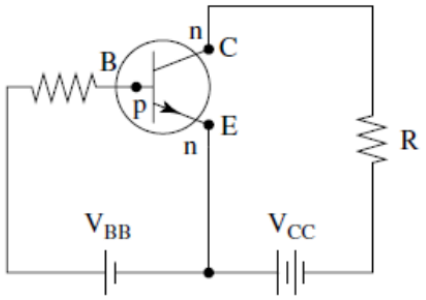
$$\beta_{dc} = \frac{\alpha_{dc}}{1 - \alpha_{dc}}$$

β_{dc} is called the base to collector current gain. This is the ratio of I_C and I_B. The value of β_{dc} varies from 25 to over 200.

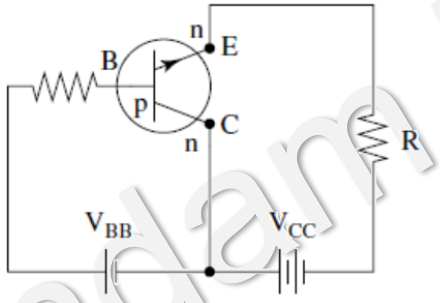
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Bipolar Junction Transistors

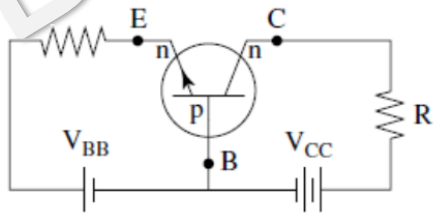
Transistor configurations: A transistor has three terminals. For the connection of input and output, one of the transistor terminals is made common, e.g., the base terminal can be made common to both input and output terminal connections. Similarly, the emitter or the collector can be connected in common configurations. Thus, we get three types of configurations, namely (i) common-base configuration; (ii) common-emitter configuration; and (iii) common-collector configuration. These three types of connections can be made for both n–p–n and p–n–p transistors.



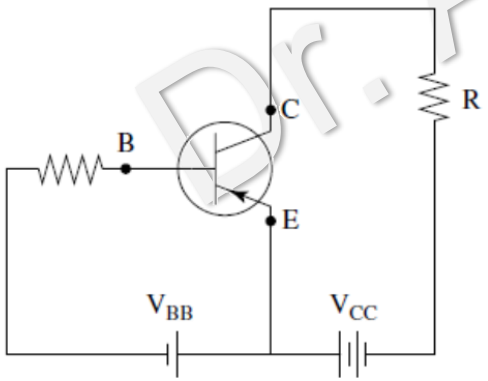
Emitter is common
n–p–n transistor



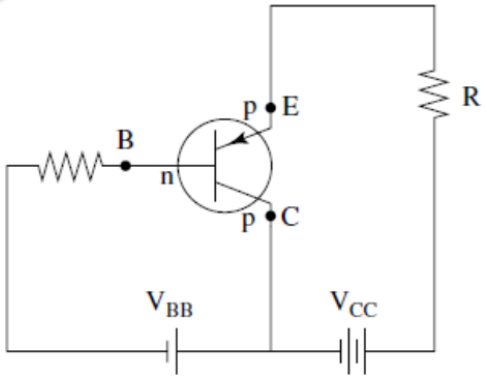
Collector is common
n–p–n transistor



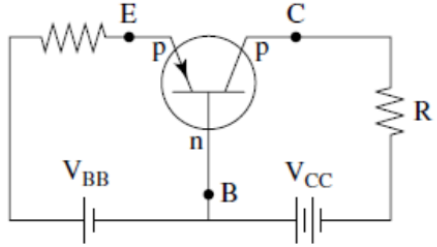
Base is common
n–p–n transistor



Emitter is common
p–n–p transistor



Collector is common
p–n–p transistor



Base is common
p–n–p transistor

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Bipolar Junction Transistors

Common-emitter transistor characteristics: In this p–n–p transistor, input voltage is applied between the base and the emitter terminals. The output is taken from the collector and the emitter terminals.

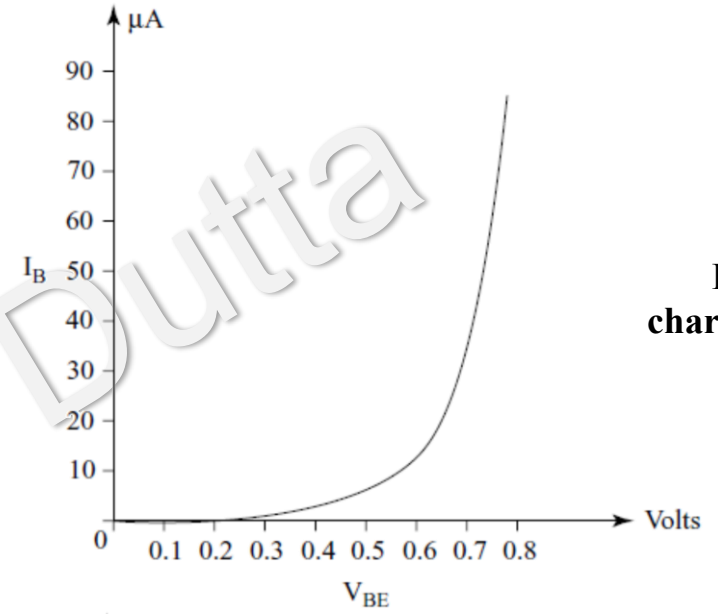
The input characteristic is drawn between V_{BE} and I_B. V_{CE} is kept constant at a value. By changing V_{BE}, current I_B is recorded. The characteristic is similar to the characteristic of a forward-biased p–n junction. The value of I_B is very small and is of the order of several microamperes only.

The output characteristics are drawn between I_C and V_{CE} keeping I_B constant at different values. For each value of I_B, V_{CE} is adjusted in steps and the values of I_C are recorded. The values of I_C are plotted against V_{CE} for each value of I_B.

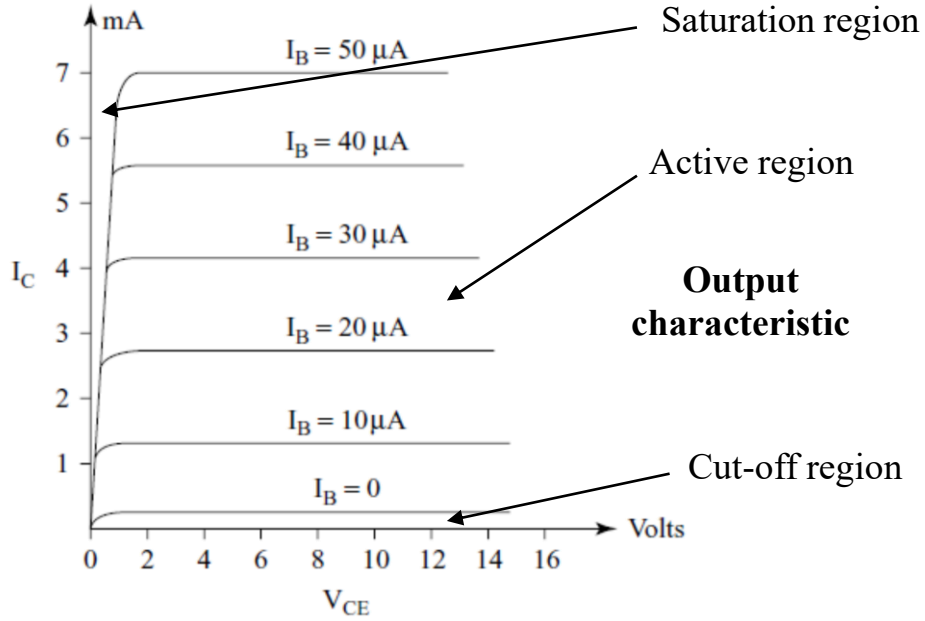
For a small change in base current (in μA) there is a large change (in mA) in collector current and emitter current. The current gain from the base to the collector can be stated as

$$\beta_{dc} = \frac{\Delta I_C}{\Delta I_B}$$

Thus, the transistor can be used as a current-amplification device. The voltage gain, A_V of the transistor is defined as the ratio of output voltage to input voltage.



Input characteristic



Output characteristic

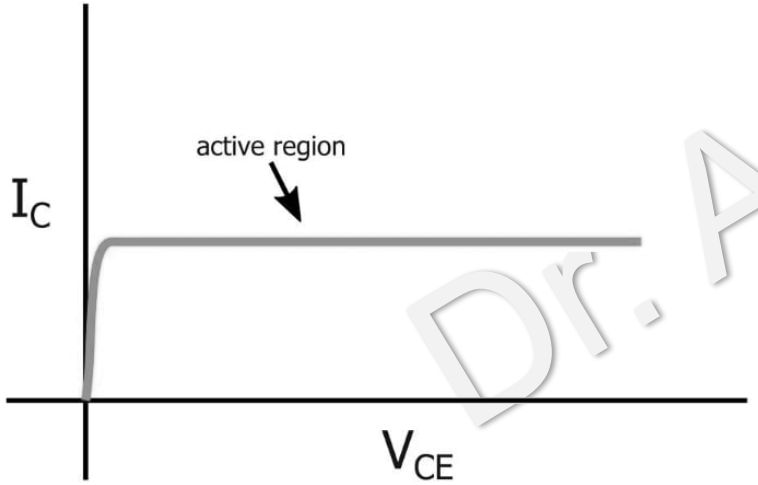
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Bipolar Junction Transistors

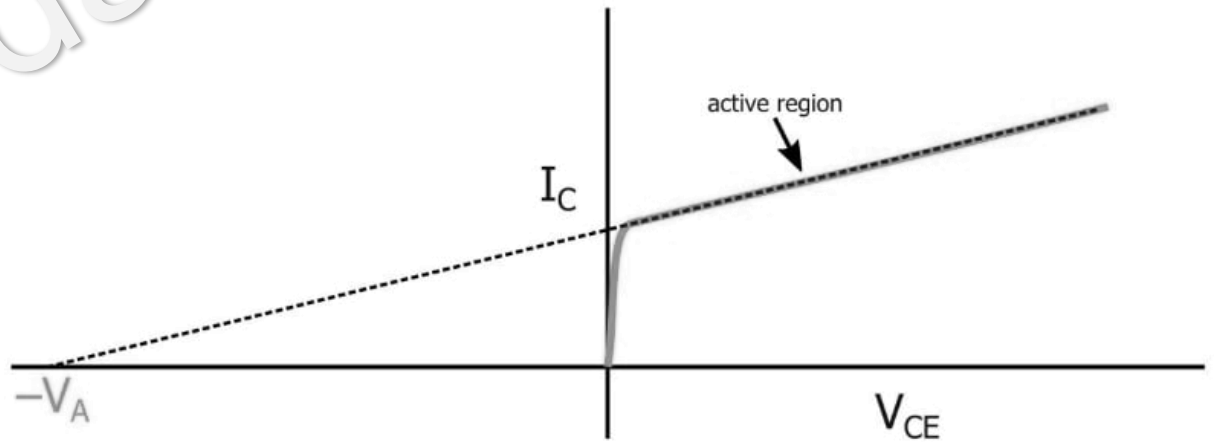
Influence of early effect in BJT characteristics: For an ideal BJT, the output characteristic would be a straight and flat line, indicating that the collector voltage has no effect on the gain.

However, if we incorporate the influence of early effect, the characteristic is still a straight line, but no longer flat. Rather, it has an upward slope, indicating that the output voltage affects the gain, because for a given I_B the collector current will be higher if the collector voltage is higher. Tracing this line back to the point at which it crosses the horizontal axis gives the Early voltage, V_A.

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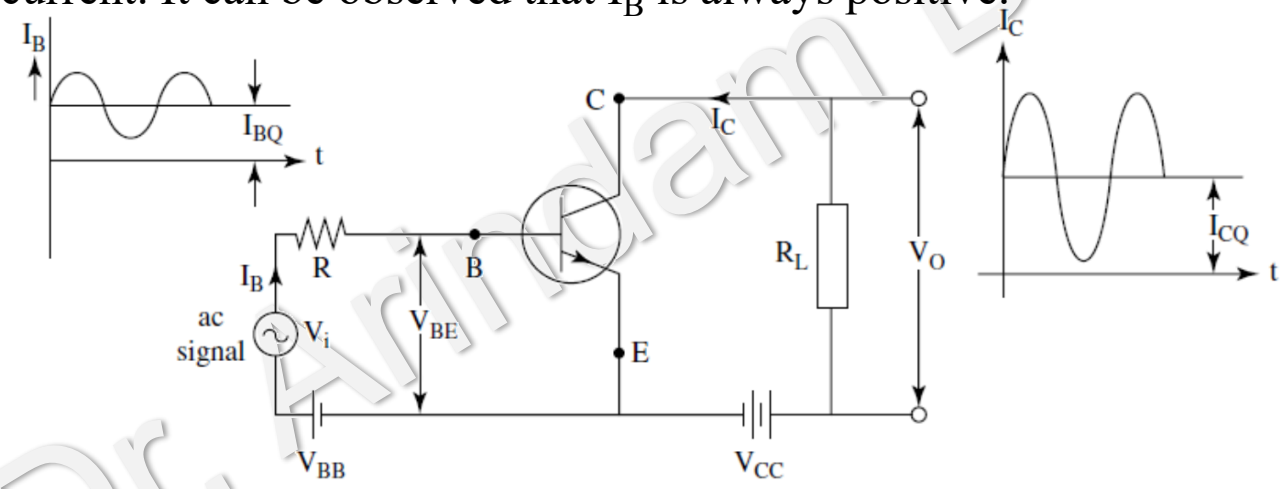
Without early effect



With early effect

Bipolar Junction Transistors

n-p-n common-emitter BJT as amplifier: The ac signal which is to be amplified is connected to the base circuit. The output is taken across a resistance, R_L in the collector circuit. The base circuit dc voltage, V_{BB} is such that the base will always remain positive irrespective of the magnitude of the input ac signal. The voltage, V_{BE} is the summation of dc voltage V_{BB} and the ac input signal, V_i. The dc voltage, V_{BB} is the bias voltage. The magnitude of V_{BB} must be higher than the maximum value of the input signal. Then, only the base will always remain positively biased in both half cycles of the input voltage. When the ac signal is applied, this becomes superimposed on the battery voltage, V_{BB} and the base current I_B will flow, which is the sum of the dc base current and the ac current. It can be observed that I_B is always positive.



During the positive half cycle of the input signal, the dc and ac voltages are added up and as such the base current is highly positive. During the negative half cycle, ac voltage is subtracted from the dc voltage. The net voltage is low but positive. The base current now will be positive but low. Because of the large variation in the base current there will be a large variation in the collector current, which will flow through the load resistance. An amplified output voltage is thus available across the load. The amplified collector ac current is superimposed on the dc current, I_{CQ} which will flow through the collector when the ac input signal is not applied. It is the current when the base current is I_{BQ}.

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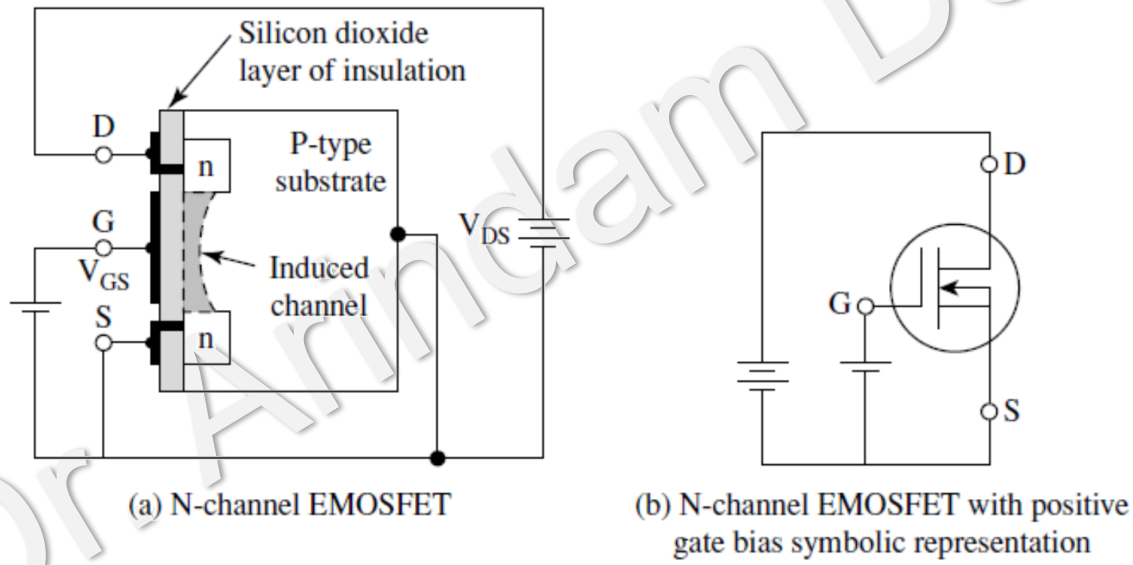
Metal Oxide Semiconductor Field Effect Transistor – MOSFET

- Field effect transistor (FET) is a voltage operated device having virtually no requirement of any input current. FETs are used in amplifier and switching circuits.
- MOSFET is a type of FET and is one of the most widely used devices incorporated in integrated circuits (IC).
- It is a three terminal device, the three-terminals are drain, source, and the gate.
- A MOSFET is also known as insulated gate field effect transistor (IGFET) as the gate of the MOSFET is insulated from the channel between the drain and the source.
- A MOSFET is commercially very popular because the MOS devices are suitable for very large-scale integration (VLSI).
- MOSFET can be n-channel-type or p-channel-type based on the channel formed between the source and the drain.
- MOSFET, either n-channel or p-channel type can further be categorized into
    - enhancement MOSFET (EMOSFET), or
    - depletion MOSFET (DEMOSFET)

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Metal Oxide Semiconductor Field Effect Transistor – MOSFET

EMOSFET: In an n-channel EMOSFET two blocks of heavily doped n-type material are diffused into a p-type substrate or a base (a substrate is a surface or material on which a device is made or developed/grown). The terminals for external connection are brought out. The whole of the surface is coated with a layer of silicon dioxide. A thin metal plate is placed on the surface of the silicon dioxide layer wherefrom the gate terminal connection is taken. In fact, the metal plate itself will function as the gate of the MOSFET. The substrate is a highly resistive material.



(a) N-channel EMOSFET

(b) N-channel EMOSFET with positive gate bias symbolic representation

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No continuous channel exists between the source and the drain. The two n-blocks forming the drain and the source, makes two p–n junctions with the p-type substrate. When a positive voltage, V_{DS} is applied between the drain and the source terminals, the p–n junction close to the drain (drain-source) is reverse biased.

Metal Oxide Semiconductor Field Effect Transistor – MOSFET

Since one of the p–n junctions is reverse biased between terminals D and S, only a negligible small current will flow between the drain and the source due to the minority charge carriers. Virtually no continuous channel exists between the drain and the source with the result that virtually no current flows between the drain and the source without any gate voltage applied.

When a positive voltage, V_{GS} is applied on the gate, the gate voltage will induce a channel by pulling the minority charge carriers of the p-type substrate towards the gate. If the gate voltage is increased more and more, negatively charged minority carriers, i.e., the electrons in this case, from the p-type substrate will get collected in the region between the drain and the source. These electrons cannot cross the SiO_2 layer, and hence constitute an n-type-induced channel between the drain and the source.

A minimum or threshold gate voltage, V_{GST} is required before drain current will flow. The magnitude of the current between the drain and the source will depend on the gate potential. Thus, the magnitude of drain current can be controlled by changing the gate voltage. The conductivity of the channel is enhanced due to the gate voltage.

With no gate voltage, the device is non-conducting, i.e., off. When the device is to be switched on, a gate voltage higher than the threshold voltage has to be applied. This type of MOSFET is therefore suitable for switching operations.

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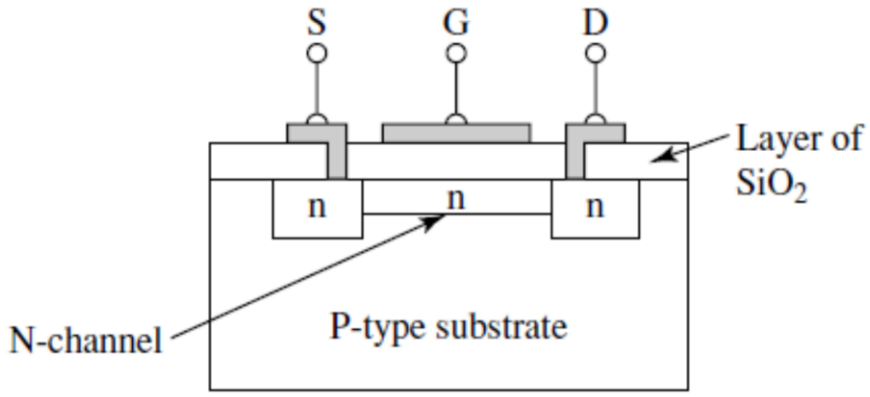
Metal Oxide Semiconductor Field Effect Transistor – MOSFET

DEMOSFET: The region between the drain and the source on the top surface of the p-type substrate is called the channel. The channel is produced by diffusing an n-type impurity material between the drain and source terminals before the insulating SiO_2 layer is deposited.

When a positive voltage V_{DS} is applied between the drain and the source, a drain current I_D flows with no gate voltage applied, i.e., when $V_{GS} = 0$.

A capacitor exists between the metal plate of the gate and the channel having the SiO_2 layer as the dielectric between the two plates. When a negative voltage is applied to the gate, one plate of the capacitor is made negative, the channel being the other plate will have positive charges induced in it. This will deplete the channel majority charge carriers, i.e., the electrons, and hence the channel conductivity will decrease. The depletion of the majority charge carriers justifies its name as depletion MOSFET. The drain current will get reduced if V_{GS} is made more negative.

A depletion MOSFET can also be made to operate in enhancement mode by applying a positive gate voltage. When a positive voltage is applied to the gate due to capacitor action, negative charges will get induced in the channel thereby increasing its conductivity. As a result, drain current will increase.

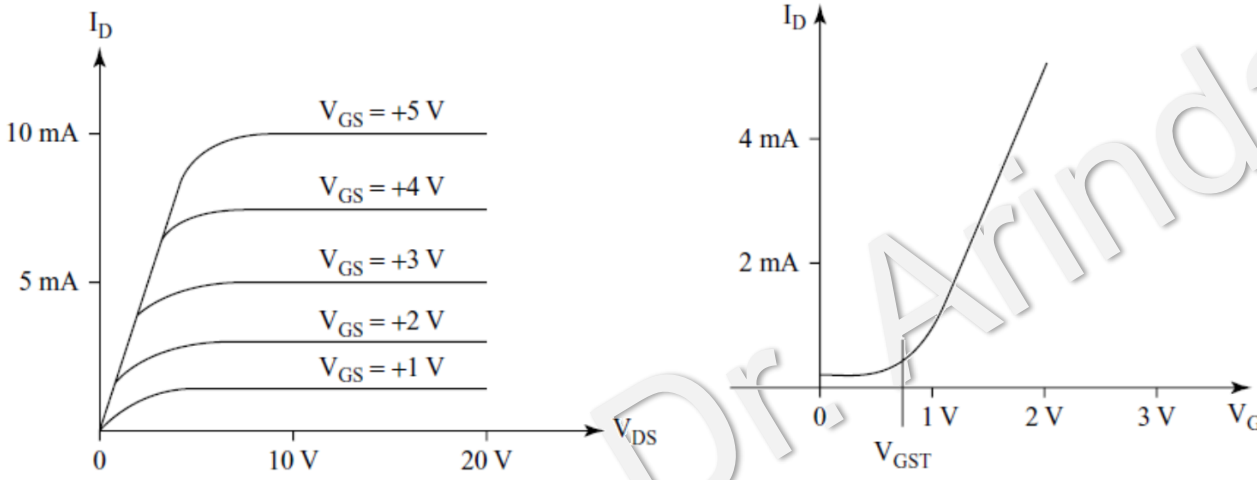


N-type DEMOSFET

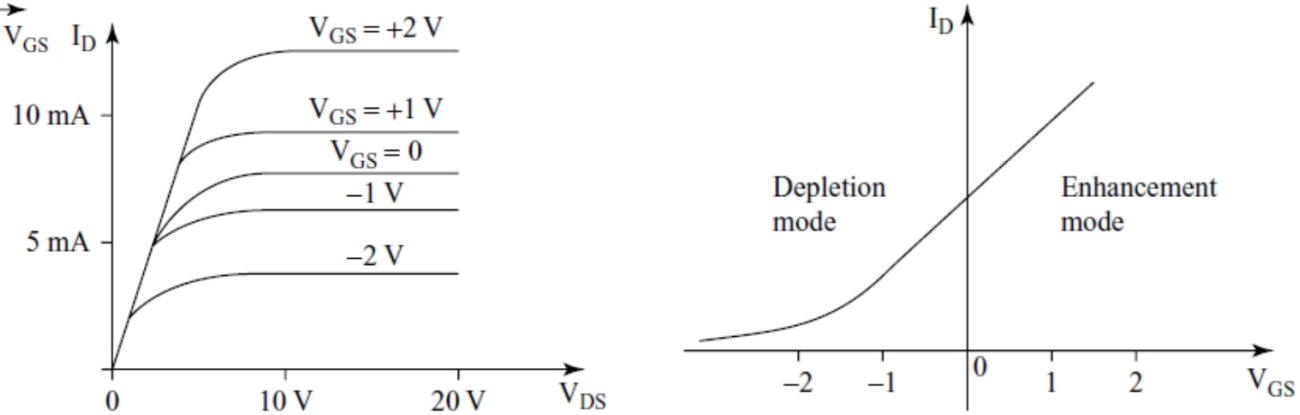
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Metal Oxide Semiconductor Field Effect Transistor – MOSFET

Static Characteristics: Figure (a) shows the typical drain characteristics of an n-channel EMOSFET. Each characteristic curve shows the variation of drain current, I_D with drain voltage V_{DS} for a fixed gate to source voltage, i.e., V_{GS}. The forward transfer characteristic (Figure (b)) is the relation between I_D and V_{GS}. Figures (c) and (d) show the drain and transfer characteristics for a DEMOSFET. In the transfer characteristic the two modes of operation of the MOSFET for negative and positive gate voltage, respectively have been shown. A MOSFET has very high input resistance of the order of 10^{10} Ω or so. MOSFETs should be handled very carefully because the very thin layer of SiO_2 is susceptible to high voltages and can be easily punctured.



(a) Typical drain characteristic of an EMOSFET; (b) transfer characteristic



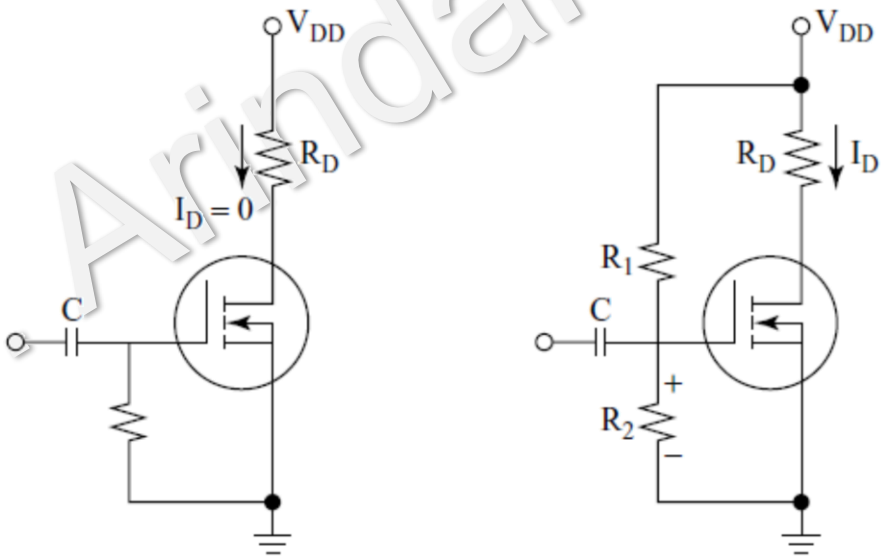
(c) Drain characteristic of an DEMOSFET; (d) transfer characteristic

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Metal Oxide Semiconductor Field Effect Transistor – MOSFET

Applications: MOSFETs are used in switching circuits, as phase shift oscillator, as an amplifier, etc.

The figures below show a capacitor-coupled MOSFET switching circuit. As shown in Fig. (a), no gate voltage is applied to the device i.e., $V_{GS} = 0$. The device is off. A positive gate voltage higher than the threshold voltage is required to turn the device on. Through the voltage divider R_1 and R_2 as in Fig. (b) a positive voltage when applied, the circuit is switched on. The device can be set to be on for a desired level of drain current, by determining the V_{GS} from the transfer characteristic provided by the manufacturer. To make sure that the device gets switched off, the V_{GS} must be brought below the minimum threshold voltage, i.e., V_{GST} .



(a) $V_{GS} = 0$, Circuit is off

(b) Positive V_{GS} applied, circuit is on

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Differences between BJT and MOSFET

Parameter	BJT	MOSFET
Terminals	• Three terminals: emitter, base, and collector	• Four terminals: source, drain, gate, and substrate
Charge carriers	• Both electrons and holes act as charge carriers	• Either electrons or holes act as charge carriers depending on the type of channel between source and drain
Polarity	• Bipolar	• Unipolar
Controlling quantity	• Current controlled	• Voltage controlled
Input impedance	• Low input impedance	• Relatively high input impedance
Power consumption	• Consumes more power than MOSFET	• Power consumed is less than BJT
Applications	• Preferred for low current applications as amplifiers, oscillators, and electronic switches	• Suitable for high power applications in power supplies, etc.

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